

FUNCTION F(x)

RETURN x^3

END FUNCTION

FUNCTION simpson13(a, b, n)

IF n is odd

n = n + 1

END IF

del = (b - a) / n

x = a + del

sum1 = 0

FOR j FROM 1 TO n-1 STEP 2

x = x + 2*del

sum1 = sum1 + F(x)

END FOR

y = a + 2*del

sum2 = 0

FOR j FROM 1 TO n-2 STEP 2

y = y + 2*del

sum2 = sum2 + F(y)

END FOR

s = (1/3)*del*(F(a) + F(b) + 4*sum1 + 2*sum2)

RETURN s

END FUNCTION

```
FUNCTION main
```

```
PRINT "a, b & n:"
```

```
READ a, b, n
```

```
s = simpson13(a, b, n)
```

```
PRINT "Integral =", s
```

```
END main
```